CHAPTER 12

# Respiration in Plants

## Glycolysis & Fermentation

**1.** Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R: *(2023)*

**Assertion (A):** ATP is used at two steps in glycolysis.

**Reason (R):** First ATP is used in converting glucose into glucose-6-phosphate and second ATP is used in conversion of fructose-6- phosphate into fructose-1-6-diphosphate.

In the light of the above statements, choose the correct answer from the options given below is true,

a. A is false but R is true.
b. Both A and R are true and R is the correct explanation of A
c. Both A and R are true but R is NOT the correct explanation of A.
d. A is true but R is false.

**2.** What is the net gain of ATP when each molecule of glucose is converted to two molecules of pyruvic acid? *(2022)*

a. Eight b. Four c. Six d. Two

**3.** What amount of energy is released from glucose during lactic acid fermentation? *(2022)*

a. Less than 7% b. Approximately 15%
c. More than 18% d. About 10%

**4.** Conversion of glucose to glucose-6-phosphate, the first irreversible reaction of glycolysis, is catalysed by *(2019)*

a. Aldolase b. Hexokinase
c. Enolase d. Phosphofructokinase

**5.** In which one of the following processes $CO_2$ is not released? *(2014)*

a. Lactate fermentation
b. Aerobic respiration in plants
c. Aerobic respiration in animals
d. Alcoholic fermentation

## Aerobic Respiration

**6.** Match **List-I** with **List-II**: *(2024 Re)*

| List-I | | List-II | |
|---|---|---|---|
| (A) | ETS Complex I | (I) | NADH Dehydrogenase |
| (B) | ETS Complex II | (II) | Cytochrome $bC_1$ |
| (C) | ETS Complex III | (III) | Cytochrome C oxidase |
| (D) | ETS Complex IV | (IV) | Succinate Dehydrogenase |

Choose the **correct** answer from the options given below:

a. A-IV, B-I, C-III, D-II b. A-I, B-IV, C-II, D-III
c. A-III, B-I, C-IV, D-II d. A-I, B-II, C-IV, D-III

**7.** Which of the following are **correct** about cellular respiration? *(2024 Re)*

A. Cellular respiration is the breaking of C-C bonds of complex organic molecules by oxidation.
B. The entire cellular respiration takes place in Mitochondria.
C. Fermentation takes place under anaerobic condition in germinating seeds.
D. The fate of pyruvate formed during glycolysis depends on the type of organism also.
E. Water is formed during respiration as a result of $O_2$ accepting electrons and getting reduced.

Choose the **correct** answer from the options given below:

a. A, C, D, E only b. A, B, E only
c. A, B, C, E only d. B, C, D, E only

**8.** Identify the step in tricarboxylic acid cycle, which does not involve oxidation of substrate. *(2024)*

a. Succinyl-CoA $\rightarrow$ Succinic acid
b. Isocitrate $\rightarrow$ $\alpha$-ketoglutaric acid
c. Malic acid $\rightarrow$ Oxaloacetic acid
d. Succinic acid $\rightarrow$ Malic acid

**9.** Match List-I with List-II *(2024)*

| List-I | | List-II | |
|---|---|---|---|
| (A) | Citric acid cycle | (I) | Cytoplasm |
| (B) | Glycolysis | (II) | Mitochondrial matrix |
| (C) | Electron transport | (III) | Intermembrane space of mitochondria |
| (D) | Proton gradient | (IV) | Inner mitochondrial membrane |

Choose the correct answer from the option given below:

a. A-(III), B-(IV), C-(I), D-(II)
b. A-(IV), B-(III), C-(II), D-(I)
c. A-(I), B-(II), C-(III), D-(IV)
d. A-(II), B-(I), C-(IV), D-(III)

**10.** Match List-I with List-II *(2023)*

| List-I | | List-II | |
|---|---|---|---|
| (A) | Oxidative decarboxylation | (I) | Citrate synthase |
| (B) | Glycolysis | (II) | Pyruvate |
| (C) | Oxidative phosphorylation | (III) | Electron transport system |
| (D) | Tricarboxylic acid cycle | (IV) | EMP pathway |

Choose the correct answer from the options given below:

a. A-II, B-IV, C-III, D-I b. A-III, B-IV, C-II, D-I
c. A-II, B-IV, C-I, D-III d. A-III, B-I, C-II, D-IV

**11.** How many times decarboxylation occurs during each TCA cycle? *(2023-Manipur)*
a. Thrice b. Many c. Once d. Twice

**12.** The 5-C compound formed during TCA cycle is: *(2022 Re)*
a. Fumaric acid b. α-ketoglutaric acid
c. Oxalo succinic acid d. Succinic acid

**13.** The number of time(s) decarboxylation of isocitrate occurs during single TCA cycle is: *(2022 Re)*
a. Four b. One c. Two d. Three

**14.** Which of the following statements is incorrect? *(2021)*
a. In ETC (Electron Transport Chain), one molecule of NADH + $H^+$ gives rise to 2 ATP molecules, and one $FADH_2$ gives rise to 3 ATP molecules.
b. ATP is synthesized through complex V.
c. Oxidation - reduction reactions produce proton gradient in respiration,
d. During aerobic respiration, role of oxygen is limited to the terminal stage.

**15.** The number of substrate level phosphorylations in one turn of citric acid cycle is: *(2020)*
a. One b. Two c. Three d. Zero

**16.** Pyruvate dehydrogenase activity during aerobic respiration requires: *(2020-Covid)*
a. Iron b. Cobalt c. Magnesium d. Calcium

**17.** What is the role of $NAD^+$ in cellular respiration? *(2018)*
a. It functions as an enzyme
b. It functions as an electron carrier
c. It is a nucleotide source for ATP synthesis
d. It is the final electron acceptor for anaerobic respiration

**18.** Which of these statements is incorrect? *(2018)*
a. Enzymes of TCA cycle are present in mitochondrial matrix.
b. Glycolysis occurs in cytosol.
c. Glycolysis operates as long as it is supplied withNAD that can pick up hydrogen atoms.
d. Oxidative phosphorylation takes place in outer mitochondrial membrane.

**19.** Which statement is wrong for Krebs' cycle? *(2017-Delhi)*
a. There are three points in the cycle where $NAD^+$ is reduced to NADH + $H^+$
b. There is one point in the cycle where $FAD^+$ is reduced to $FADH_2$
c. During conversion of succinyl CoA to succinic acid, a molecule of GTP is synthesised
d. The cycle starts with condensation of acetyl group (acetyl CoA) with pyruvic acid to yield citric acid

**20.** Oxidative phosphorylation is: *(2016 - II)*
a. Addition of phosphate group to ATP.
b. Formation of ATP by energy released from electrons removed during substrate oxidation.
c. Formation of ATP by transfer of phosphate group from a substrate to ADP.
d. Oxidation of phosphate group in ATP.

**21.** Cytochromes are found in: *(2015)*
a. Cristae of mitochondria
b. Lysosomes
c. Matrix of mitochondria
d. Outer wall of mitochondria

## Respiratory Balance Sheet & Amphibolic Pathway

**22.** Fatty acids are connected with the respiratory pathway through: *(2023-Manipur)*
a. Acetyl CoA
b. α-Ketoglutaric acid
c. Dihydroxy acetone phosphate
d. Pyruvic acid

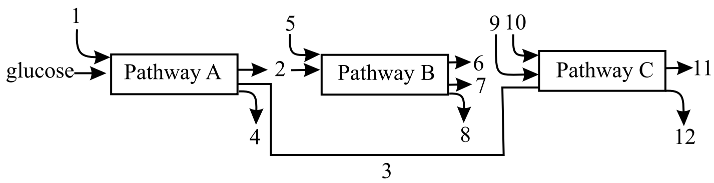
**23.** Match List-I with List-II: *(2022 Re)*

| List-I | | List-II | |
|---|---|---|---|
| (A) | Porins | (I) | Pink coloured nodules |
| (B) | leg haemoglobin | (II) | Lumen of thylakoid |
| (C) | $H^+$ accumulation | (III) | Amphibolic pathway |
| (D) | Respiration | (IV) | Huge pores in outer membrane of mitochondria |

Choose the correct answer from the options given below:
a. A-II, B-IV, C-I, D-III b. A-II, B-I, C-IV, D-III
c. A-IV, B-I, C-II, D-III d. A-III, B-IV, C-II, D-I

**24.** Which of the following biomolecules is common to respiration-mediated breakdown of fats, carbohydrates and proteins? *(2016 - II, 2003)*
a. Pyruvic acid b. Acetyl CoA
c. Glucose-6-phosphate d. Fructose 1,6-bisphosphate

**25.** The three boxes in this diagram represent the three major biosynthetic pathways in aerobic respiration. Arrows represent net reactants or products: *(2013)*




Arrows numbered 4, 8, and 12 can all be
a. $FAD^+$ or $FADH_2$ b. NADH
c. ATP d. $H_2O$

## Respiratory Quotient

**26.** Respiratory Quotient (RQ) value of tripalmitin is *(2019)*
a. 0.9 b. 0.7 c. 0.07 d. 0.09

## Answer Key

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **1.** (b) | **2.** (d) | **3.** (a) | **4.** (b) | **5.** (a) | **6.** (b) | **7.** (a) | **8.** (a) | **9.** (d) | **10.** (a) |
| **11.** (d) | **12.** (b) | **13.** (b) | **14.** (a) | **15.** (a) | **16.** (c) | **17.** (b) | **18.** (d) | **19.** (d) | **20.** (b) |
| **21.** (a) | **22.** (a) | **23.** (c) | **24.** (b) | **25.** (c) | **26.** (b) | | | | |